Preserving Visual Evidence in Compressed Multimodal Reasoning

Research Proposal

Abstract

Short multimodal explanations must preserve the visual evidence supporting their answers. We propose image-conditioned sufficiency supervision for concise traces retaining necessary premises and valid deductions. The study uses one 7B answering model, annotated diagrams, human-checked image pairs, and a separate judge of explanatory support. Evidence and style rewards will be compared on identical rewritten data at matched compute and achieved reasoning-token budgets. Original-trace and length-preserving rewrite controls test rewriting effects. We hypothesize improved premise retention and accuracy under compression.

1 Introduction

Chain-of-thought prompting makes intermediate reasoning explicit, while reinforcement learning can improve reasoning performance [1, 2]. In visual mathematics, a short solution must retain facts read from diagrams alongside algebraic deductions. Removing "the marked angle is 40°" can leave an otherwise plausible calculation unsupported. Whether compression preferentially removes these visual premises remains an empirical question.

SketchThinker-R1 [3] learns concise traces using a style judge that sees neither the image nor the question. Consequently, its style supervision does not establish preservation of answer-critical visual evidence. Its Table 2 reports MATH-Vision accuracy of 25.3% for SketchThinker-R1-3B versus 26.9% for Vanilla-R1-3B, a decrease of 1.6 percentage points. This comparison motivates testing accuracy across compression budgets.

We ask whether evidence supervision improves the accuracy–length tradeoff beyond style supervision: (RQ1) are necessary visual premises omitted more often than necessary nonvisual steps? (RQ2) does sufficiency supervision improve accuracy and premise retention at matched average reasoning-token budgets, including harder items? (RQ3) does any advantage persist when rewritten data and training compute are controlled? The contributions are a sufficiency criterion, an annotated intervention set with a validated judge, and a controlled comparison of correctness and explanation sufficiency.

2 Related Work

Concise reasoning and budget control. Sketch-of-Thought uses constrained reasoning paradigms to shorten outputs [4]; s1 controls test-time reasoning through budget forcing [5]. These motivate our budget sweeps. SketchThinker-R1 combines cold-start rewriting, style-judge training, and reinforcement learning [3]. Its default reward is $0.5R_{\text{accuracy}} + 0.4R_{\text{format}} + 0.1R_{\text{style}}$, rather than style alone. A fixed style-reward coefficient does not impose a common token budget: trace lengths can vary across problems. The paper also examines reward schedules. Our controls test the contribution of cold-start rewriting.

Verification and visual dependence. Process supervision evaluates intermediate reasoning rather than only final answers [6]. Evidence that explanations can misrepresent a model's reasons motivates separating explanatory support from causal faithfulness [7]. MathVerse varies the distribution of information between question text and diagrams [8]; MathVista and MATH-Vision assess visual mathematical reasoning [9, 10]. We evaluate explicit visual premises during compression using checked labels and controlled edits. Qwen2.5-VL supplies the multimodal backbone [11], and Group Relative Policy Optimization (GRPO), introduced in DeepSeekMath [12], supplies the policy-training procedure.

3 Proposed Methodology

Task and operational criterion. Given image I and question q, policy π_θ produces trace z and answer a. The trace must conclude with a, supported by its derivation. The answer field contains only the normalized final answer; all preceding explanation counts toward reasoning length $L(z)$, with target budget B. Reviewers construct a nonempty set $\mathscr{G}(I,q)$ of validated admissible derivation graphs with visual premises, textual premises, and deductions. Visual nodes carry region references or measurable relations. A unit is necessary relative to a derivation if its removal breaks support for the answer.

For each $G \in \mathscr{G}(I,q)$, $P_{V,G}(z)$ and $P_{N,G}(z)$ are the fractions of required visual premises and nonvisual units explicitly stated and correct. Units are atomic semantic propositions under a fixed annotation rubric. Empty categories have coverage 1; category-specific summaries exclude them, report their counts, and macro-average over remaining problems. Let $D(z) \in \{0,1\}$ indicate a valid derivation of the returned answer from stated premises, question facts, and declared elementary rules; $D_G(z)$ applies this check to G. Define

$$S(z;I,q,\mathscr{G}) = \max_{G \in \mathscr{G}(I,q)} \mathbf{1}[P_{V,G}(z) = 1]\,\mathbf{1}[P_{N,G}(z) = 1]\,D_G(z). \tag{1}$$

The image verifies stated facts; it cannot supply missing premises. Thus "$x = 180 - 40 - 60 = 80$" fails Eq. 1 if the angle readings occur only in the image. Redundant statements can be deleted when a complete admissible derivation remains.

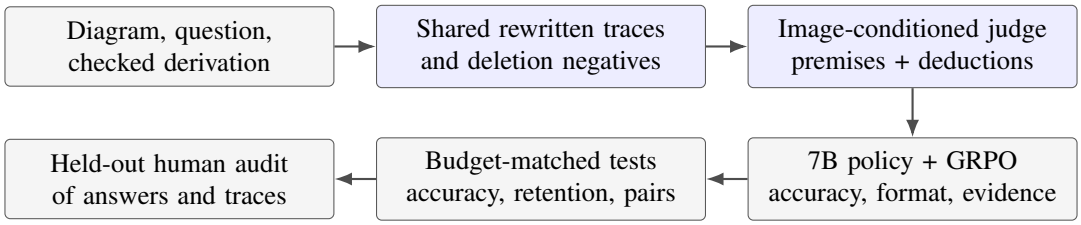


Figure 1: Proposed pipeline. Training annotations supervise the judge, development annotations calibrate it, and held-out human annotations evaluate the judge and generated traces.

Judge architecture and training. A separate Qwen2.5-VL-3B judge will receive I,q,z and predict structured premise-grounding and deduction-validity labels, followed by a binary sufficiency label. Supervised token cross-entropy trains both outputs. The score $s_\phi(I,q,z)$ is the normalized probability of the sufficient label over the two allowed label tokens, calibrated on development data. Reference graphs supply training labels; inference uses I,q,z alone. Training uses supported sketches, single-premise deletions, incorrect readings, invalid deductions, and redundant-step deletions, matched in length and format. Negatives include correct answers with missing evidence.

Two annotators independently label the audited subset; grounding and deduction agreement are reported before adjudication. Development data set threshold τ for at most 10% false acceptance and at least 70% sufficient-trace recall. Missing either target triggers more labeling before policy training. Held-out precision, recall, calibration error, and false acceptance assess transfer.

Compression and policy optimization. All principal reward arms will start from the same Qwen2.5-VL-7B-Instruct checkpoint, fine-tuned on identical compressed traces from one frozen rewriter. The policy learns to generate sketches directly. Figure 1 summarizes the workflow; Eq. 2 defines the reward:

$$R = 0.5\mathbf{1}[a = a^*] + 0.4F(z,a) + 0.1E_\phi(I,q,z) - \lambda \max(0, L(z)/B - 1), \tag{2}$$

where a^* is the checked answer, F checks response format, and $E_\phi = \mathbf{1}[s_\phi \geq \tau]$. GRPO uses four rollouts per group to save compute versus the five in SketchThinker-R1 [3], advantages $A_i = (r_i - \bar{r})/\max(\sigma_r, \varepsilon)$

with fixed $\varepsilon > 0$ and $A_i = 0$ for equal-reward groups, and a KL penalty to the cold-start policy. The style arm replaces E_ϕ with the source's binary text-only judgment $R_{\text{style}} \in \{0,1\}$; both auxiliary rewards have support $\{0,1\}$ and coefficient 0.1. A probability-valued evidence reward is a separate ablation; a zero-component arm sets $E_\phi = 0$. All use the same length penalty and development-only coefficient search. Budgets $B \in \{64,128,256,512\}$ set targets; evaluation matches achieved lengths. The judge is used during training only; deployment requires the answering model alone.

4 Experimental Plan

Data and leakage controls. We plan 2,400 programmatically rendered geometry and bar-chart problems, split into 1,600 training, 300 development, and 500 test problems by template family. Generators supply reference facts and solutions. Humans review 600 training, 100 development, and 200 test problems and their candidate traces; test premise scores use adjudicated annotations. From test originals we build 200 answer-changing and 100 nuisance-edit image pairs, checking both members for validity and preservation of unrelated facts. Related images, templates, and traces share a split.

Table 1 summarizes the evaluation. Newly trained components exclude public benchmark items from training and model selection. Released checkpoints are compared separately by known benchmark exposure. SketchRL-1K includes MathVista questions, so SketchThinker-R1 results on MathVista require an item-level overlap audit.

Table 1: Planned datasets, evaluation measures, and comparisons.

Dataset / role	Metrics	Baselines / comparison
Rendered diagrams; training, development, test	Answer accuracy; visual NPR; nonvisual retention; human sufficiency	Shared-data style reward, evidence reward, zero component
Checked image pairs; test only	Both-correct rate; evidence consistency; nuisance invariance	Same three reward arms; judge without image
MathVista testmini; MATH-Vision test; MathVerse testmini [9, 10, 8]	Official answer scoring; reasoning tokens; end-to-end latency; difficulty groups where available	Long reasoning; direct answer; constrained budget; Sketch-of-Thought prompting [4]; SketchThinker-R1-7B [3] (separate checkpoint comparison)

Measurements and falsifiable tests. RQ1 measures omission within retained derivations, separately from incorrect grounding. For $C \in \{V,N\}$, let U_C contain necessary units explicit and correct in the original trace, and $M_u(z)$ indicate an explicit semantic mention of u, regardless of correctness. Write z_c for the compressed trace. For nonempty U_C, $d_C^{\text{omit}} = 1 - |U_C|^{-1} \sum_{u \in U_C} M_u(z_c)$ measures omission. Category rates exclude empty sets, report their counts, and macro-average over remaining problems. Incorrect grounding, invalid steps, and adjudicated valid changes of derivation are reported separately. RQ2 will measure answer accuracy, visual necessary-premise retention (NPR), and human-rated sufficiency separately. Paired correctness requires both answers correct; evidence consistency requires each regenerated trace to support its answer with premises true in its own image. Valid relational wording can apply to both images. Nuisance edits test answer invariance.

Fair comparisons and uncertainty. Development operating points will match achieved average reasoning-token counts within 5%. Unmatched points are reported separately. The public-test evaluation will use frozen settings and one answer per item, with parser failures counted as incorrect. Latency runs from receipt of prepared input through final-answer completion, including image encoding, image/text prefill, reasoning-trace decoding, and final-answer decoding, on fixed hardware at batch size one after warm-up. MATH-Vision's supplied difficulty levels and annotation-defined derivation-depth groups will support prespecified stratified analysis. Three policy-training seeds and paired bootstrap 95% confidence intervals will quantify

uncertainty, resampling original problem families together with their edits and versions and reporting seed variation. A pilot will estimate the detectable effect.

For RQ3, the primary evidence-versus-style contrast holds rewritten examples, cold-start weights, decoding, and hyperparameter-search allowance fixed. Training will be capped by measured compute, including rollout generation and judge inference, rather than equal update counts alone. We will report processed text/image tokens, estimated FLOPs, accelerator-hours, and separate preprocessing, cold-start, judge-training, and RL costs. The compressed-trace zero-component arm, original-trace control, and same-rewriter length-preserving control use Eq. 2 with $E_\phi = 0$ and identical accuracy, format, and length terms at each operating point. These contrasts vary cold-start traces under a fixed policy-training objective; content audits track rewriting changes. Image-ablated judges and removal of the evidence component will test which supervision matters.

5 Expected Results & Timeline

We predict greater omission of necessary visual premises than nonvisual steps under text-only compression, and higher NPR and accuracy from evidence supervision at matched budgets, especially on deeper derivations. Null accuracy gains, poor held-out judge precision, or gains lost under compute controls would weaken the predicted advantage. Retention gains alone would support explanation quality.

The six-month plan (Table 2) requires four 80GB accelerators and parameter-efficient policy tuning. With $K \leq 4$ separately trained budgets, three arms and three seeds require $9K$ principal GRPO runs, up to 36. The month-one pilot will estimate GPU-hours per run, including rollouts and judge inference, and enumerate search, control, and ablation runs. Four accelerators supply 2,880 GPU-hours per 30-day month. At usable fraction u, month-four training must satisfy $H_{\text{principal}} + H_{\text{overhead}} \leq 2,880u$: at $K = 4$, at most $80u$ GPU-hours per principal run before shared overhead. We will report pilot estimates and reduce K if projected costs exceed capacity, retaining three seeds and the principal contrast. Month-five controls receive a separate GPU-hour allocation. Blinded human audits, validated alternative proofs, and separate public-benchmark results address judge errors, annotation ambiguity, and transfer risk. Outputs include code, rubrics, permitted data and edits, calibration records, and accuracy–cost curves, including null results.

Table 2: Month-by-month milestones and decision points.

Month	Activities and deliverables
1	Reproduce baselines; pilot compute; freeze splits, scoring, and annotation rubric.
2	Generate and annotate diagrams; adjudicate pairs; pilot deletion-rate analysis.
3	Train and calibrate judges; audit negatives; check acceptance and recall targets.
4	Train principal reward arms with three seeds; freeze budget operating points.
5	Run controls, ablations, and held-out tests; analyze uncertainty and difficulty.
6	Complete human audit, failure analysis, reproducibility package, and report.

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